

Technology Stack

Date: 28 June 2026 **Team ID:** **Project Name:** Rising Waters – AI Flood Prediction System
Maximum Marks: 2 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build your solution. Provide a brief justification for each technology.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to build a responsive and interactive user interface for entering flood parameters and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, and integrates the machine learning model.
3	Database / Data Storage	SQLite	Stores prediction history and application data in a lightweight local database.
4	Cloud / Hosting / Deployment	Local Flask Server (localhost)	Used to run and test the application during development. <i>(If you deploy later, you can replace this with Render.)</i>
5	Version Control & CI/CD	Git, GitHub	Used for version control, project collaboration, and source code management.
6	Third-Party APIs / Other Tools	XGBoost, Pandas, NumPy, Scikit-learn, VS Code	Used for machine learning, data preprocessing, model training, and application development.

